- Pharmacology
- Influenza Antiviral Agents
- SARS-CoV-2 Antiviral Agents

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After completing the preparation materials, students should be able to:

- Give examples of how each of the pharmacokinetic properties of the antiviral agents contributes to patient care.
- Describe the mechanisms of action of each class in the context of the functions of the targeted viral biomolecules.
- Relate viral resistance to the antiviral mechanisms of action.
- Explain the goals of therapy and evidence-based rationales for the clinical selection of specific antiviral drugs.
- List the potential adverse effects and drug interactions.

Preparation Materials (links are in the CPG and on the next slide)

Required
• ScholarRx Bricks Practice Questions
Suggested additional resources:
• Video Lectures Dr. Goldstein's Notes handout Guided Reading Questions
• Textbook resources with links are listed on the next slide

SUGGESTIONS:
Use the resources that work best for you.
You do not need to study all of them.
Work through the GUIDED READING QUESTIONS with pen/pencil and paper.
Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links to Resources

Title of Brick: Antiviral Drugs >
Foundations and Frameworks https://exchange.scholarrx.com/brick/antiviral-drugs-foundations-and-frameworks
Drugs to Treat Influenza Virus and Respiratory Syncytial Virus
Link: https://exchange.scholarrx.com/brick/drugs-to-treat-influenza-virus-and-respiratory-syncytial-virus-infection
There are currently no Bricks on SARS-CoV-2 antiviral agents.
Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024: Chapter 49: Antiviral Agents > Anti-influenza Agents
https://accessmedicine.mhmedical.com.nyit.idm.oclc.org/content.aspx?bookid=3382&sectionid=281755260
Access Medicine Katzung's Basic & Clinical Pharmacology: Examination & Board Review, 14e, 2024; Chapter 49: Antiviral Agents > Anti-influenza Agents
https://accessmedicine.mhmedical.com.nyit.idm.oclc.org/content.aspx?bookid=3461&sectionid=285598965
LWW Health Library Medical Education: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 34: Antiviral Agents > Treatment of Respiratory Infections
https://premiumbasicsciences.lwwhealthlibrary.com.nyit.idm.oclc.org/content.aspx?sectionid=253329261&bookid=3222

Key points: Influenza Antiviral Agents

- Influenza antiviral agents are important adjuncts to influenza vaccine in the control of influenza.
- They are used for treatment of acute uncomplicated influenza, post-exposure prophylaxis in patients at risk of influenza complications, and pre-exposure prophylaxis in widespread influenza outbreaks for unvaccinated individuals at very high risk of developing complications, such as immunocompromised patients.
- When initiated within 48 hours of the onset of symptoms, treatment can lessen symptoms, shorten illness duration by 1 or 2 days, decrease duration of viral shedding, and may reduce the risk of lower respiratory tract complications and hospitalizations.
- The different agents have various routes of administration, half-lives, and indications for use.
- Prophylaxis does not eliminate risk of influenza; susceptibility returns when the drug is removed. Adjunctive vaccine should be administered.

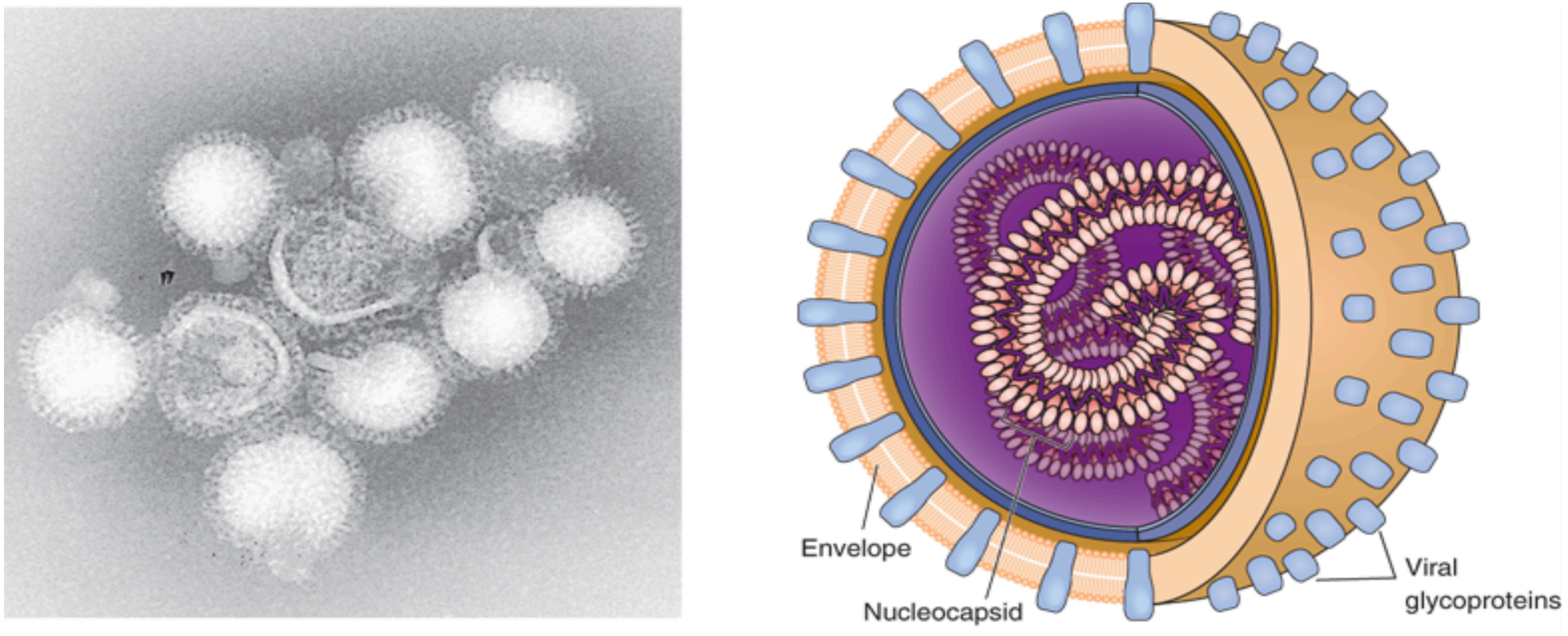
Key points

- The neuraminidase inhibitors (NAIs), oseltamivir, zanamivir, and peramivir, prevent viral cleavage of sialic acid residues from viral proteins, which prevents the release of new influenza A and B virions. Oseltamivir and zanamivir are neuraminic acid analogs. Peramivir was designed to have high affinity binding to the neuraminidase active site. Viral resistance to the NAIs is rare.
- **Oseltamivir** is the preferred agent for treatment and prophylaxis of influenza in outpatients and hospitalized patients. It is typically taken orally twice daily x5 days for treatment. It can cause nausea and vomiting. Taking with food may improve tolerance.
- **Zanamivir** for oral inhalation twice daily x5 days (treatment) is an alternative to oseltamivir. It can cause bronchospasm and should be avoided in patients with underlying respiratory disease. Allergic reactions are reported.
- **Peramivir** is an alternative to oseltamivir for treatment only. It is not indicated for prophylaxis. It has a long half-life and is given intravenously x1 dose for outpatients and hospitalized patients. Adverse effects include hyperglycemia, skin reactions, hypersensitivity reactions.
- NAIs are associated with transient neuropsychiatric events – confusion, delirium, hallucinations.
- Oseltamivir is the preferred drug for pregnant individuals. Zanamivir and peramivir are alternatives. All NAIs may be administered to pediatric patients (drug-specific age recommendations).

Key points

- **Baloxavir** is an influenza cap-dependent endonuclease (PA protein) inhibitor effective against influenza A and B. Point mutations in viral PA protein confer resistance. It is administered orally x1 dose for treatment and post-exposure prophylaxis. Administration along with polyvalent cations (foods, oral supplements, and bile acid sequestrants) should be avoided. Baloxavir is generally well tolerated. Baloxavir is not currently recommended in pregnancy due to lack of safety and efficacy data.
- **Adamantanes, amatadine and rimantadine**, prevent viral uncoating and replication by blocking the M2 protein ion channel of influenza A, but not influenza B. Adamantanes are not recommended for influenza treatment or prophylaxis because of high-level resistance by A/H1N1 and A/H3N2 viruses.
- Amantadine distributes to the CNS and can cause cognitive effects, agitation, hallucinations, and others. It is used as an adjunct in Parkinson disease treatment. Rimantadine has a low incidence of CNS effects. They are both taken orally.

Influenza viruses overview



Source: Karen C. Carroll, Stephen A. Morse, Timothy Mietzner, Steve Miller: Jawetz, Melnick, & Adelberg's Medical Microbiology, 27th Edition.
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Influenza virus. A: Electron micrograph of influenza virus A/Hong Kong/1/68(H3N2). Note the pleomorphic shapes and glycoprotein projections covering particle surfaces (315,000×). (Courtesy of FA Murphy and EL Palmer.) B: Schematic view of influenza. Virus particles have segmented genomes consisting of seven or eight different RNA molecules, each coated by capsid proteins and forming helical nucleocapsids. Viral glycoproteins (hemagglutinin and neuraminidase) protrude as spikes through the lipid envelope.
(Reproduced with permission from Willey JM, Sherwood LM, Woolverton CJ: Prescott, Harley, and Klein's Microbiology, 7th ed. McGraw Hill, 2008. © The McGraw-Hill Companies, Inc.)

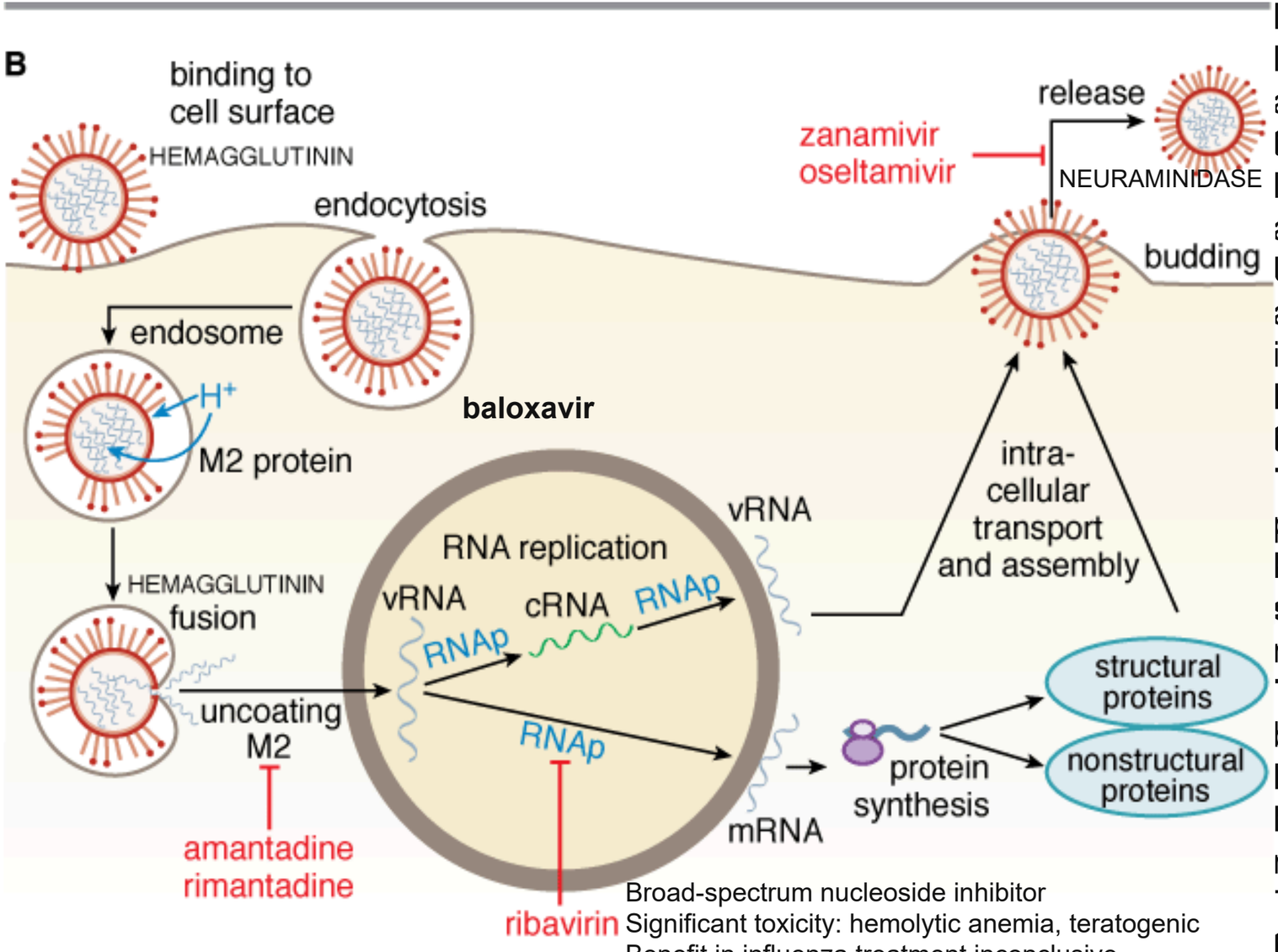
Citation: Chapter 39 Orthomyxoviruses (Influenza Viruses), Riedel S, Hobden JA, Miller S, Morse SA, Mietzner TA, Detrick B, Mitchell TG, Sakanari JA, Hotez Mejia R. *Jawetz, Melnick, & Adelberg's Medical Microbiology, 28e;* 2019, Figure 39-1. Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2629&sectionid=217775144 Accessed: August 28, 2023
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The Anti-Influenza Antiviral Drug Classes

Neuraminidase Inhibitors

Cap-Dependent Endonuclease Inhibitor

Adamantanes



Broad-spectrum nucleoside inhibitor
Significant toxicity: hemolytic anemia, teratogenic
Benefit in influenza treatment inconclusive

Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com
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Influenza virus life cycle and drug targets.
Binding: Hemagglutinin glycoproteins bind sialic acids on the surface of epithelial cell membranes of the upper respiratory tract.
Entry: Virus particles fuse with the membrane and are taken up by endocytosis.
Uncoating: The **M2 protein** of influenza virus allows an influx of hydrogen ions into the virion interior, disrupting the viral envelope. Nucleocapsids enter the cytoplasm and migrate to the nucleus.
Transcription: The endonuclease subunit (PA protein) of viral RNA polymerase (RNAp in figure = RdRp) cleaves a primer from cellular mRNA (**cap snatching**). RNAp synthesizes mRNA, which moves into the cytoplasm.
Translation: The capped mRNAs are recognized by host ribosomes and translated into viral proteins.
Nucleocapsid assembly occurs in the cytoplasm.
Final virus assembly takes place at the plasma membrane.
The antiviral agents and targets are indicated on the figure in red (non-black) font.

Citation: Chapter 62 Antiviral Agents (Nonretroviral), Brunton LL, Hilal-Dandan R, Knollmann BC. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e;* 2017.
Available at: https://accessmedicine.mhmedical.com/content.aspx?bookid=2189&sectionid=172486085 Accessed: November 13, 2021
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Slide adapted by LG

Therapeutic uses of the anti-influenza class of drugs

Treatment of acute uncomplicated influenza
Start within 48 hours of the onset of signs / symptoms
Treatment can:
- Lessen symptoms
- Shorten illness duration by 1 or 2 days
- Decrease duration of viral shedding
- May reduce the risk of lower respiratory tract complications and hospitalizations in adults and children compared to placebo or no treatment
- oseltamivir, zanamivir, peramivir, baloxavir

Prophylaxis: Vaccination

Post-exposure Prophylaxis: Administer within 48 hours of exposure
- Individuals at high risk for severe or complicated influenza
- Healthy unvaccinated people who have close contact with individuals at high risk of complication of influenza
- oseltamivir, zanamivir, or baloxavir

Pre-exposure Prophylaxis:
During widespread outbreaks, consider for unvax. pts at very high risk of developing complications
oseltamivir, zanamivir

Neuraminidase Inhibitors (NAIs)

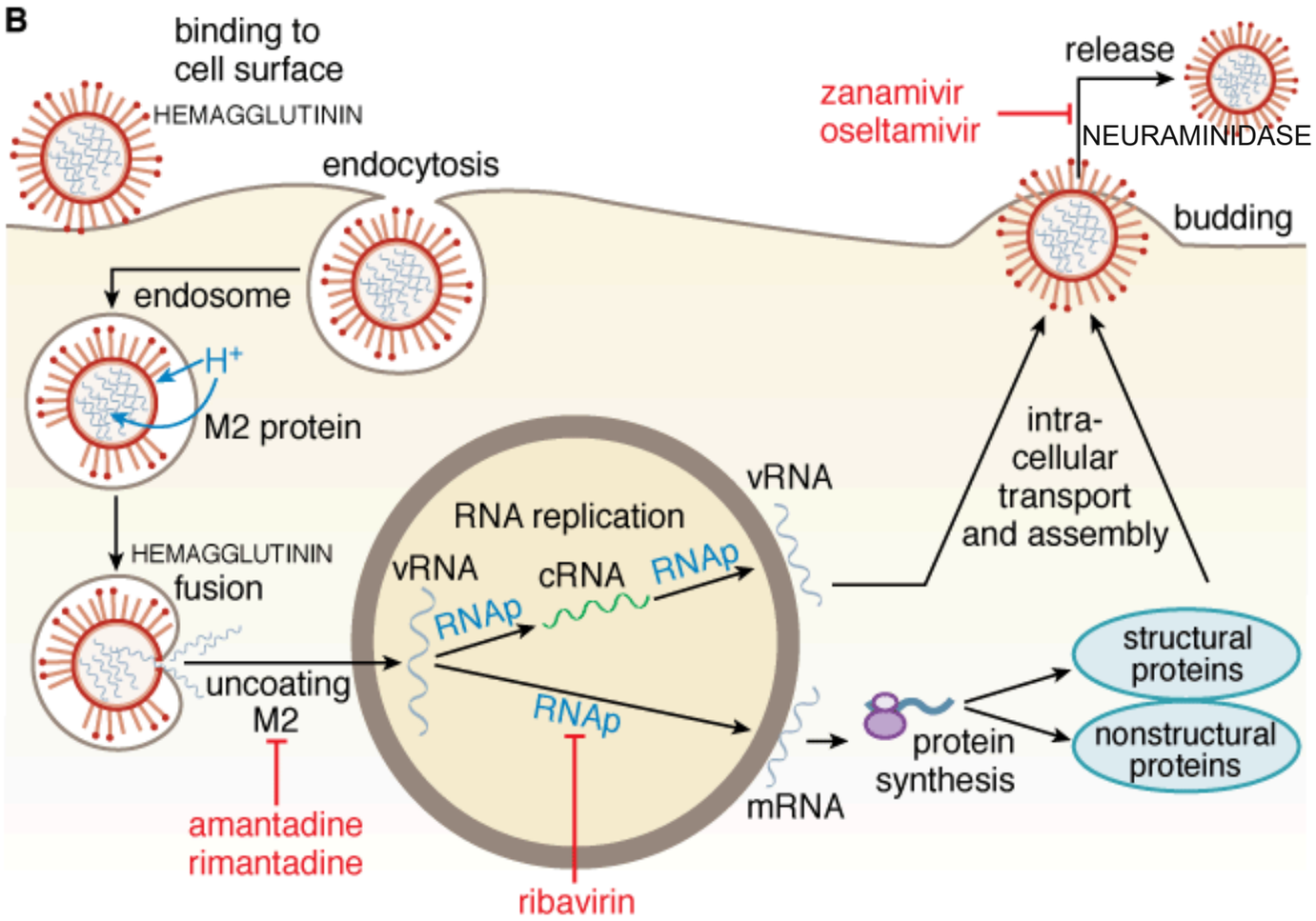
- ->Oseltamivir
- ->Zanamivir
- ->Peramivir

Neuraminidase Inhibitors Pharmacokinetic Properties

Takeaways: “Favorable” PK profile		Oseltamivir	Zanamivir	Peramivir
Adherence to therapy	A	Oral prodrug Tx: 2x/d x5 days typically PPx: 1x/d x7-14d after exposure	Oral inhalation (powder) Low systemic absorption Tx: 2x/d x5 days PPx: as for oseltamivir	IV x 1 dose Hospitalized patients: Tx:1x/d x5-10 days (*oseltamivir preferred*)
	D	Low to moderate plasma protein binding free drug to tissues to site of action		
No metabolism-related drug interactions	M	Esterase hydrolysis to active carboxylate Bioavailability ~75%	Minimal metabolism	Minimal metabolism
	E	Renal, OAT1/3 secretion of parent and active metabolite	Urine, unchanged (less than 8% of dose)	Renal, unchanged drug
Renal dosing		Dose adjustment for renal impairment		Dose adjustment for renal impairment
Dose frequency	t½	6-10 h (active metabolite)	Up to 5 h (serum)	20 h
		All NAIs may be used in pediatrics, adolescents, adults, and pregnancy.		

Neuraminidase Inhibitors
Mechanism of action

- drug binds to neuraminidase
-
- induces conformational change that prevents cleavage of the
- sialic acid molecule
-
- aggregation (clumping) of viruses to each other and to membrane
-
- interferes with
- release of viral progeny
- **Activity:** Flu virus A and B
- **Resistance:** rare



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition*
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Goodman & Gilman Figure 58-1B

Therapeutic uses of NAIs

drug of choice	alternatives to oseltamivir	
Oseltamivir	Zanamivir	Peramivir
Treatment and prophylaxis of infants, children, adoles cents, and adults outpatients and hospitalized patients	Treatment, outpatients: children ≥7 years old and adults Prophylaxis in children ≥5 years and adults • Not recommended in patients with underlying respiratory disease (eg asthma, COPD)	Treatment, outpatients: children ≥2 years old and adults Treatment, hospitalized pts: Alternative to oseltamivir if the patient cannot tolerate or absorb oseltamivir Not used for prophylaxis

- Oseltamivir is the preferred drug for the treatment and prophylaxis of influenza in pregnancy.

Zanamivir and peramavir are alternatives if oseltamivir is not tolerated.

- Neuraminidase Inhibitors Potential Adverse Effects
- These agents are generally well tolerated.

Oseltamivir	Zanamivir	Peramivir
Most common: • Nausea and vomiting May be taken with food to improve tolerance • Headache • Serious skin reactions	• Headache • Bronchospasm risk in patients with underlying airways disease • Sinusitis; dizziness; ear, nose, throat infections • Allergic reactions: oropharyngeal or facial edema Not recommended for patients with underlying respiratory disease (eg asthma, COPD)	• Diarrhea or constipation • Hyperglycemia • Serious skin reactions • Hypersensitivity reactions
NAIs Neuropsychiatric events, rare, transient (post-marketing surveillance) Confusion, delirium, hallucinations potentially with behavioral or neurologic changes Occur primarily in children and adolescents. The cause has not been established.		

- **Cap-Dependent Endonuclease Inhibitor**
- ->Baloxavir marboxil

Baloxavir marboxil Pharmacokinetics

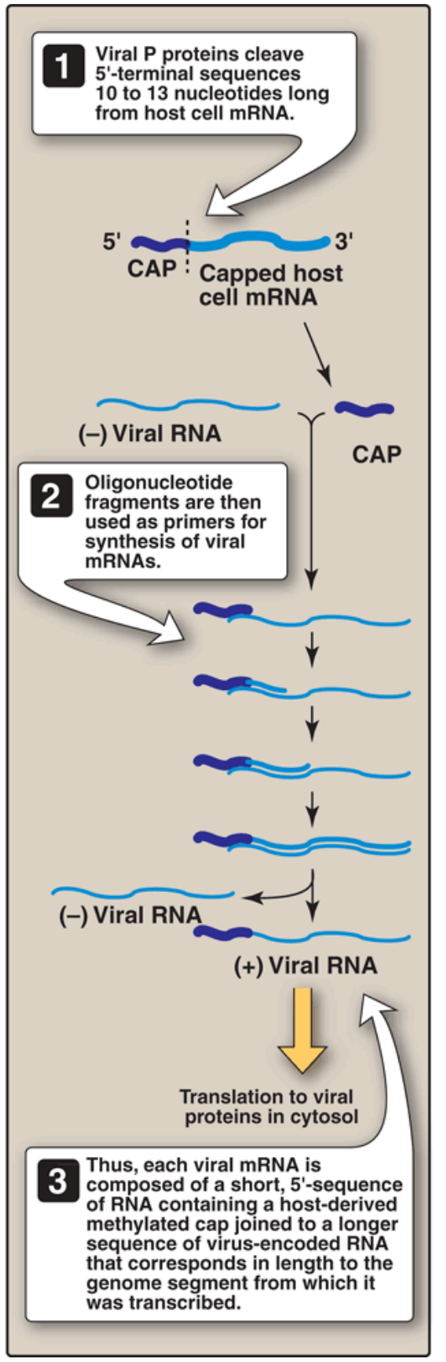
only one dose needed	A	Oral prodrug **Single dose** within 48 hours of symptom onset reduces symptoms by approximately 1-2 days
low potential for metabolism, transport DDIs	D	Very large Vd (1,180 L) minor P-glycoprotein substrate Vd helped inform the manufacturer's dosing recommendations. • As weight increases, exposure to baloxavir decreases higher dose for patients with a higher BMI
	M	UGT1A3 conversion to active metabolite, baloxavir (minor CYP3A4)
renal dose adjust. likely not needed only one dose needed	E t½	Feces (80%) Urine (15%; 3% as baloxavir) ~80 hours

LWW Health Library

From: **Chapter 29 Negative-strand RNA Viruses, Figure 29.11**

Lippincott's Illustrated Reviews: Microbiology, 3e, 2012

“Cap snatching” by influenza virus prior to viral mRNA translation.
As with other negative-strand RNA viruses, the viral genomic RNA serves as the template for synthesis of viral mRNAs. Each of the eight genome segments directs the synthesis of one positive-strand mRNA. However, influenza virus is distinguished from the other negative-strand viruses in that, although the virions contain proteins that can transcribe mRNAs, these enzymes lack the ability to cap and methylate viral mRNAs.
The synthesis of influenza virus mRNA, therefore, begins with “cap snatching,” in which the viral P protein cleaves 5’-terminal sequences (10 to 13 nucleotides long) from nascent host cellular Pol II transcripts that had been previously synthesized, capped, and methylated in the nucleus. The oligonucleotide fragments are then used as primers for the synthesis of the viral mRNAs. Thus, each viral mRNA is composed of a short, 5’-sequence of RNA containing a host-derived methylated cap joined to a longer sequence of virus-encoded RNA that corresponds in length to the genome segment from which it was transcribed.



Baloxavir Effects

- Activity: Effective against influenza A and B
- Resistance may emerge – point mutations in PA protein
- Therapeutic use:
  - Acute uncomplicated influenza and post-exposure prophylaxis in patients ≥12 years at high risk of developing complications
  - Children 5 years who are otherwise healthy
  - NOT recommended in pregnancy – lack of safety and efficacy data
- Adverse effects: Generally well tolerated
  Low incidence of side effects:
  - headache
  - diarrhea
  - nasopharyngitis
- Drug interactions: polyvalent cations
  - Chelate baloxavir absorption of, thus, exposure of the body to the drug
  - Avoid calcium-containing foods and cation-containing supplements

Adamantanes

->Amantadine

->Rimantadine

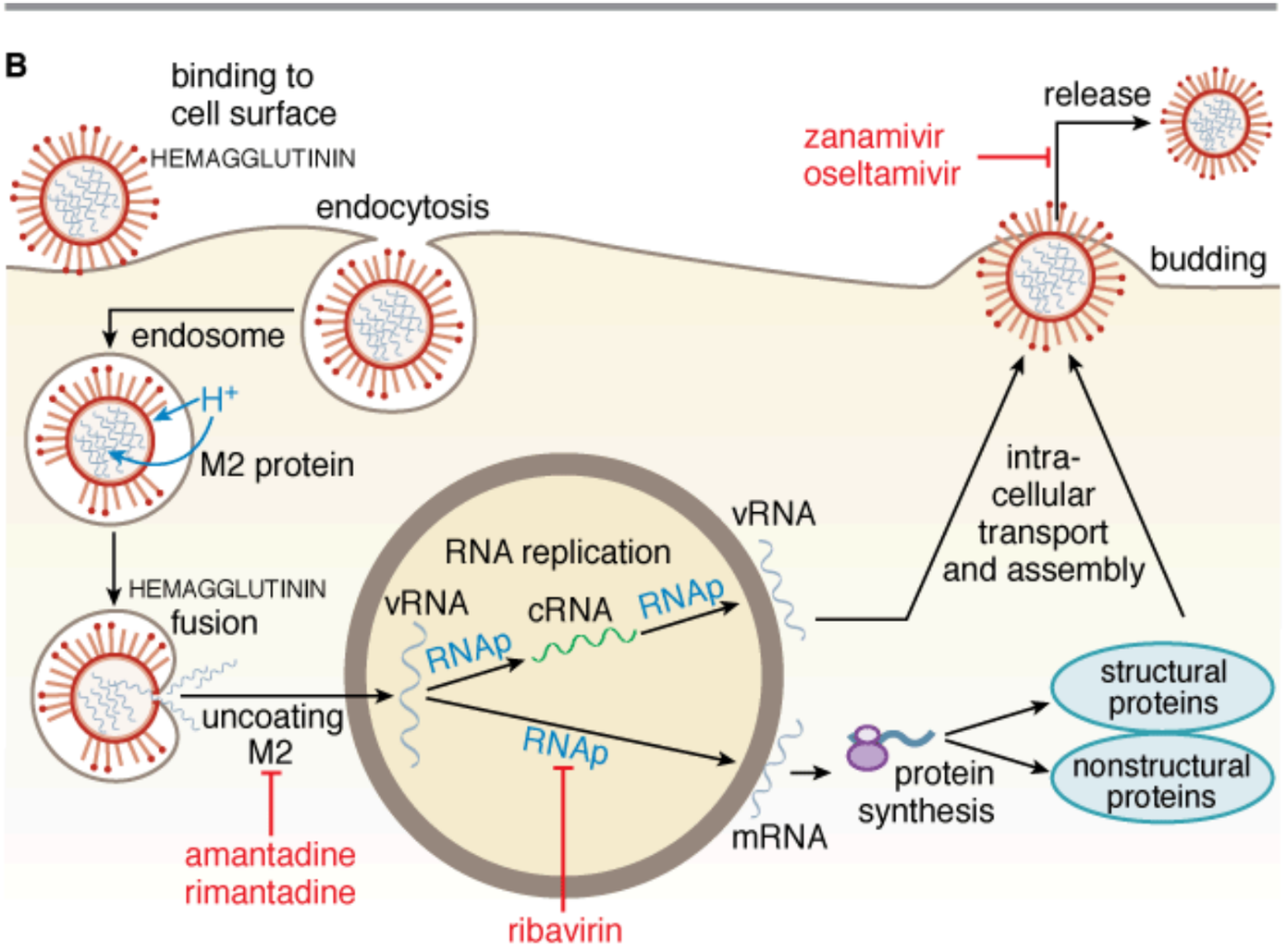
Adamantanes are no longer used for influenza treatment or prophylaxis due to high-level resistance and activity against influenza A only.

Adamantanes Pharmacokinetic Properties

	Amantadine	Rimantadine
A	Oral Bioavailability ~90%	Oral Well absorbed
D	Distributes to CNS ***Elderly are more sensitive to amantadine's CNS adverse effects.***	Low CNS distribution Moderate protein binding
M	Minimal	Extensive hepatic hydroxylation, conjugation, glucuronidation
E	Urine, ~90% unchanged	Urine, metabolites
	Dosage adjustment recommended in severe renal or hepatic impairment.	
t½	12-18 hours: young adults 29 hours: healthy older adults 8 days: end stage renal disease (not dialyzable)	13-65 hours: adults 20-65 hours: elderly

Adamantanes Mechanism of action and Resistance

- M2 transmembrane protein interference
-
- block proton ion channel
-
- inhibit uncoating
- and viral assembly
- **Activity:** influenza A only
- **Resistance:**
- **high prevalence**
- develops rapidly



Source: Brunton LL, Chabner BA, Knollmann BC: *Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12th Edition:* www.accessmedicine.com
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M2 protein = proton ion channel

Adamantanes Adverse Effects

Amantadine: CNS side effects
difficulty concentrating, agitation, insomnia, confusion, hallucinations
High concentrations serious neurotoxicities possible
Elderly are ***especially vulnerable*** to the CNS effects.

The neurotoxic effects of amantadine appear to be increased by concomitant ingestion of antihistamines and psychotropic or anticholinergic drugs, especially in the elderly.

Rimantadine: CNS side effects occur less frequently

Summary of Influenza Antiviral Agents

Prophylaxis and Treatment of Acute Uncomplicated Influenza

Neuraminidase Inhibitors

Competitive inhibition of the enzyme viral clumping interferes with release of virus progeny

Oseltamivir preferred agent	Zanamivir	Peramivir
Oral	Oral inhalation	I.V.
All ages	Pts 7 yrs	Pts 6 mon
BID x5d	BID x5d	1 dose
Nausea	Bronchospasm	Skin reactions

Neuropsychiatric adverse effects (rare)
Confusion, delirium, hallucinations
May be associated with behavioral or neurologic changes

Cap-dependent Endonuclease Inhibitor

Baloxavir inhibits host mRNA “cap snatching” inhibits initiation of RNA synthesis

Baloxavir marboxil

Oral

Pts 5 yrs

1 dose

Diarrhea, Hypersensitivity

Chelates polyvalent cations

Amantadine, Rimantadine: M2 proton ion channel inhibitors prevent uncoating/assembly, influenza A only
Not recommended – Resistance H1N1, H3N2

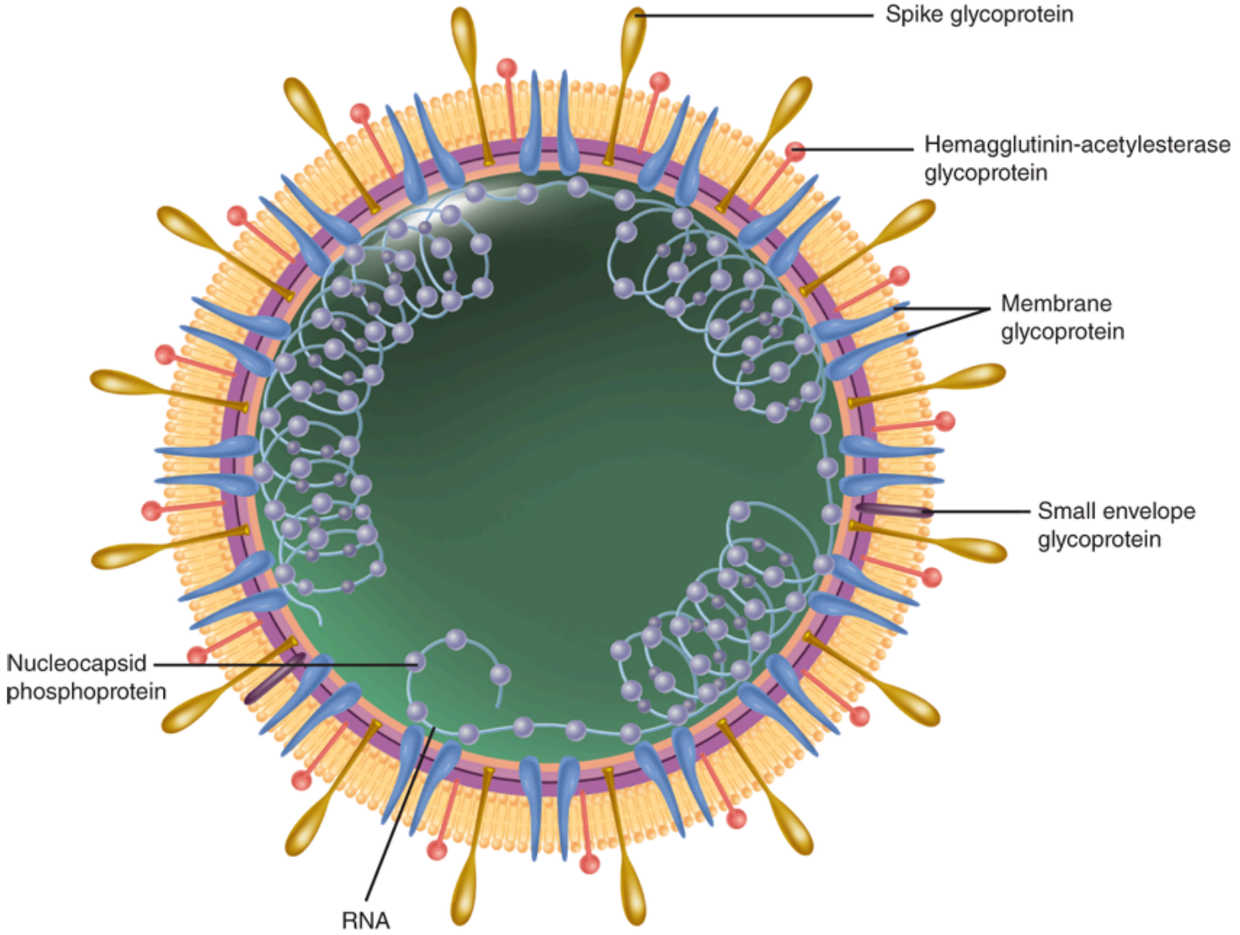
- COVID-19
- SARS-CoV-2 Antiviral Agents

Key points: COVID-19 Antiviral Therapy

- To determine the appropriateness of COVID-19-specific antiviral therapy, patients are evaluated for suspected or confirmed COVID-19 to assess the severity of symptoms, risk of progression to severe disease, and safety of the home setting, with shared clinical decision making with the patient.
- Outpatient treatment of acute illness is recommended for symptomatic patients with mild to moderate illness and risk factors for progression to severe disease, such as older age, immune status, COVID-19 vaccination history, or comorbidities associated with progression.
- **Nirmatrelvir-ritonavir** is the recommended agent. Nirmatrelvir is a peptidomimetic inhibitor of SARS-CoV-2 3CLPRO (same as main protease MPRO), which prevents processing of viral polyprotein precursors, thus, inhibiting viral replication. Ritonavir inhibits CYP3A4, which increases nirmatrelvir concentrations. The drug combination is initiated as soon as possible within 5 days of symptom onset and administered twice daily for five days. Studies have shown that it substantially reduces the risk of hospitalization and mortality in high-risk individuals with mild to moderate COVID-19. It is FDA approved for adults, adolescents, and children 12 years old. Renal dosing is listed in the drug monograph. There is a high potential for drug interactions.
- **Ensitrelvir** (FDA approved May 2026) is a SARS-CoV-2 3CLPRO inhibitor. It has a long half-life and does not need a PK booster.

Key points

- **Remdesivir** is an adenosine nucleotide analog activated by triphosphorylation, which competes for incorporation into viral RNA chains by RNA-dependent RNA polymerase (RdRp), resulting in delayed chain termination. Remdesivir is FDA approved for the treatment of mild to moderate COVID-19 in non-hospitalized and hospitalized infants 3 kg, children, adolescents, and adults. The doses are once-daily intravenous infusions x3 days in non-hospitalized patients and 5 days usually or until discharge for hospitalized patients (not on mechanical ventilation). The drug reduces COVID-19 hospitalizations in acute symptomatic patients with mild to moderate illness and may reduce time to recovery and risk of mechanical ventilation in hospitalized patients with severe COVID-19 not on mechanical ventilation.
- **Molnupiravir** has US Emergency Use Authorization (EUA) for outpatient treatment of mild to moderate symptomatic patients 18 years old who are at high risk of progression to severe disease. Potential benefit is likely modest. Molnupiravir did not reduce the risk of hospitalization but did reduce the time to recovery. The drug is metabolized to cytidine nucleoside analog, NHC. Triphosphorylated NHC is incorporated into SARS-CoV-2 RNA viral RdRp. When RdRp copies the RNA strand containing NHC, mutations accumulate, resulting in errors in the viral genome – lethal mutagenesis (viral error catastrophe) – inhibiting replication. The drug is initiated as soon as possible within 5 days of symptom onset and administered twice daily for five days.
- **Pemivibart** is an IgG1 antibody directed against SARS-CoV-2 spike protein for prophylaxis of COVID-19 in patients ≥12 years old weighing ≥40 kg with moderate to severe immune compromise who cannot be vaccinated or have inadequate response to vaccination (US Emergency Use Authorization).



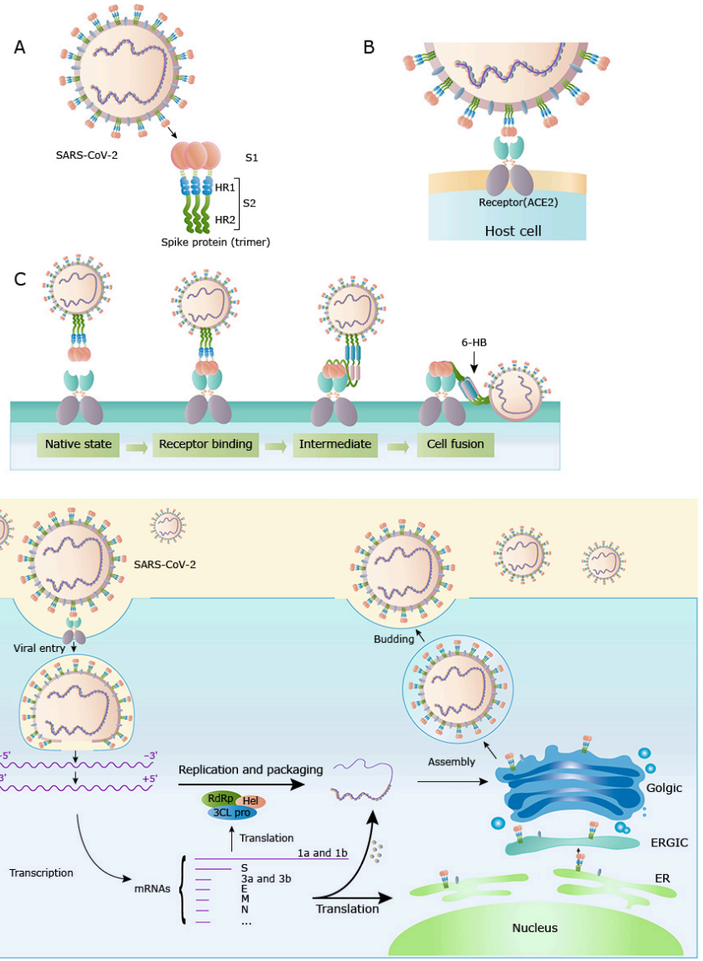
Source: Kenneth J. Ryan:
Sherris Medical Microbiology, Seventh Edition
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Virion structure of a coronavirus. Coronavirus particle is shown to contain a single-stranded, positive-sense RNA genome bound to a nucleoprotein (helical nucleocapsid) surrounded by a lipid bilayer envelope. Petal- or club-shaped spikes (spike glycoprotein) project from the surface of the envelope giving the appearance of a crown of thorns or a solar corona. There are several other surface proteins, including hemagglutinin-acetylesterase glycoprotein, membrane glycoprotein, and small envelope glycoprotein.

Citation: Chapter 9 Respiratory Viruses, Ryan KJ. *Sherris Medical Microbiology, 7e;* 2017, Figure 9-9. Available at: https://accessmedicine.mhmedical.com/content.aspx?sectionid=176082624&bookid=2268 Accessed: November 12, 2021
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SARS-CoV-2 Replication

Structure and function of the SARS-CoV-2 spike protein



(A) The schematic structure of the spike (S) protein.

(B) The spike (S) protein binds to the receptor ACE2.

(C) The binding and virus-cell fusion process mediated by the spike (S) protein.

(D) The life cycle of SARS-CoV-2 in host cells.

SARS-CoV-2: severe acute respiratory syndrome coronavirus 2.

Reprinted by permission from Macmillan Publishers Ltd: Acta Pharmacologica Sinica. Huang Y, Yang C, Xu XF, et al. Structural and functional properties of SARS-CoV-2 spike protein: potential antivirus drug development for COVID-19. Acta Pharmacol Sin 2020; 41:1141. Copyright © 2020. https://www.nature.com/aps/.

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- The primary host cell entry receptor for SARS-CoV-2 is angiotensin-converting enzyme 2 (ACE2). SARS-CoV-2 binds ACE2 through the receptor-binding domain of its spike protein and together with host factors the virus fuses with the cell's outer membrane and enters the cell by endocytosis.
- After uncoating, the viral positive-sense RNA uses the cell's machinery to produce structural proteins necessary for assembling new virus particles and to produce complementary negative-sense RNA to synthesize more positive-stranded RNA genomes. The new viruses are transported by the Golgi apparatus to the cell membrane where they are released from the host cell by exocytosis or upon cell lysis.

Variants:

Additional enzymatic interactions between SARS-CoV-2 and host cells determine specific cell tropism, which has evolved with the virus.

Following ACE2 binding, the original SARS-CoV-2 virus and some subsequent variants, including Delta, enter cells after a host protease (most often the TMPRSS2 enzyme which is expressed in high levels in lung epithelial cells) cleaves the spike protein, enabling subsequent membrane fusion and cell entry.

Omicron-lineage variants, in contrast, mostly enter host cells through a different pathway involving cathepsin-dependent endocytosis and use the TMPRSS2-mediated pathway far less efficiently. Thus, Omicron-lineage variants replicate more efficiently in the upper airways, where TMPRSS2 expression is lower than in the lungs.

SARS-CoV-2 Antiviral Agents

- ->Nirmatrelvir with Ritonavir (peptidomimetic with PK booster)
- Ensitrelvir (FDA approved May 2026)
- ->Remdesivir (adenosine nucleotide analog)
- ->Molnupiravir (cytidine nucleoside analog)
- Pemivibart (antibody directed against spike protein)

FDA approval: Nirmatrelvir with ritonavir and Remdesivir
US Emergency Use Authorization: Molnupiravir and Pemivibart

Pharmacokinetics

	Nirmatrelvir with ritonavir	**Remdesivir**	**Molnupiravir**
A	**Oral**	**Prodrug, IV only**	**Oral prodrug** of N4-hydroxycytidine (NHC)
D	Widely distributed in tissues Protein binding ~90% (ritonavir)	Widely distributed in tissues Protein binding ~90%	Widely distributed in tissues NHC minimal protein binding
M	Nirmatrelvir: **CYP3A4** but minimal metabolism when given with ritonavir	Intracellular phosphorylation to active ribonucleoside triphosphate **remdesivir-TP**	Molnupiravir: ester hydrolysis to active **NHC** Intracellular phosphorylation to active ribonucleoside triphosphate **NHC-TP**
	Ritonavir inhibits CYP3A4 increases (boosts) nirmatrelvir bioavailability and increases t½.		
E	Urine/feces **Dose adjust. renal impairment** Not recommended in severe hepatic impairment	Urine, primarily metabolites No dose adjust. renal impairment Not recommended in severe hepatic impairment	Urine (3%) as NHC No dose adjust. renal impairment
t½	Nirmatrelvir: ~6 h when given with ritonavir Ritonavir: ~6 h • **Every 12h daily** for 5 days (Pills are packaged in blister packs)	Remdesivir prodrug 1 h Active metabolite 27 h • Outpatient: **Once daily** x3 days • Inpatient: **Once daily** x5d usually	NHC: ~3 h • **Every 12 h for 5 days**

Mechanisms of Action: All these agents are effective against the ancestral virus and the other variants that cause illness.

Nirmatrelvir Protease inhibitor	**Remdesivir** RdRp inhibitor	**Molnupiravir** “Viral error catastrophe”
Inhibits 3CLPRO, 3CLPRO cleaves SARS-CoV-2 polyproteins, 1a and 1b Inhibits the release of nonstructural proteins, including proteases Inhibits viral replication	Remdesivir adenosine nucleotide intracellular phosphorylation remdesivir-triphosphosphate (TP) competes with adenosine-TP for incorporation into RNA chains by RdRp chain termination after incorporation of a few more nucleotides delayed chain termination during viral replication	**Step 1.** NHC intracellular phosphorylation active NHC-TP **Step 2.** RdRp takes up NHC-TP as a substrate instead of cytosine-TP and uridine-TP. **Step 3.** When RdRp copies the RNA strand containing NHC, more mutations are incorporated, which leads to mispairing and *accumulation of mutated RNA* eventually lethal to the virus. **Result:** lethal mutagenesis “viral error catastrophe”
Ritonavir: Pharmacokinetic enhancer CYP3A4 inhibitor		
Ensitrelvir 3CLPRO inhibitor Does not require PK booster		

RdRp: RNA-dependent RNA polymerase

Therapeutic Use: Administer as soon as possible after onset of symptoms and within 5 days of symptom onset.

	Nirmatrelvir with ritonavir	**Remdesivir**	**Molnupiravir**
Use	Mild to moderate illness Outpatients at high risk for progression to severe COVID-19 or hospitalization ***Preferred agent***	Mild to moderate illness Outpatients at high risk for progression to severe COVID-19* Hospitalized patients not on mechanical ventilation.	Mild to moderate illness Outpatients at risk for severe COVID-19 or hospitalization Alternative agent
	Adults Adolescents 12 years and 40 kg	Adults Infants 28 days, 3 kg Children and Adolescents	Adults Adolescents 18 years and 40 kg
Pregnancy	Treatment recommended if indicated.	Treatment recommended if indicated.	Not recommended. Patients who may become pregnant should use reliable contraception during therapy and for 4 days after the last dose.
	*Remdesivir may be administered only in health care settings where clinicians have immediate access to medications for treating severe infusion reactions or hypersensitivity reactions.		

Risk factors for progression to severe disease

- Older age (65 years)
- Overweight / Obesity
(adults, BMI 25 kg/m2; age 12-17, BMI 85th percentile for age/sex)
- Pregnancy
- Chronic kidney disease
- Diabetes mellitus
- Immunosuppression (immunosuppressive disease or treatment)
- Cardiovascular disease (including congenital heart disease)
- Hypertension
- Chronic lung diseases
- Sickle cell disease
- Neurodevelopmental disorders
- Other medically complex conditions (eg, genetic or metabolic syndromes, severe congenital abnormalities)
- Dependence on medically-related technologies unrelated to COVID-19 (tracheostomy, gastrostomy, positive pressure ventilation)

Efficacy

- Nirmatrelvir with ritonavir substantially reduces the risk of hospitalization and mortality in high-risk individuals with mild to moderate COVID-19.
- Remdesivir reduces COVID-19 hospitalizations in acute symptomatic patients with mild to moderate illness and may reduce time to recovery and risk of mechanical ventilation in hospitalized patients with severe COVID-19 not on mechanical ventilation.
- Molnupiravir potential benefit is likely modest. It reduces time to recovery but does not reduce the risk of hospitalization.

Note about COVID-19 Rebound: After negative antigen testing following completion of the 5-day course of nirmatrelvir with ritonavir, some patients experience return of symptoms or test positivity usually within the first 10 days. Recurrent symptoms are generally mild and retreatment is not routinely recommended.

Adverse Effects (not a complete list)

Nirmatrelvir with ritonavir	• Nausea, diarrhea • Dysgeusia (bad taste – people complain about this) • Rash, including serious skin reactions • Ritonavir: elevated liver enzymes, hypersensitivity reactions

Drug interactions – *High potential*: CYP3A4-mediated interactions

- Take a thorough medication history including prescription and OTC medications, botanical products, and potential food interactions
- Use a high-quality tool to assess specific drug interactions and potential ways to mitigate them is highly recommended.
- The following UpToDate Lexidrug Drug Interactions Tool is accessible through the NYITCOM Library

https://www-uptodate-com.nyit.idm.oclc.org/contents/search?unid=l0mV%2fPt0iyUkHS7yMWtEWIiqlXybNUsM&srcsys=EZPX

Adverse Effects (not a complete list)

Remdesivir	• Severe bradycardia (mechanism not known) • Increased serum aminotransferases, reversible; acute hepatic failure • Hypersensitivity reactions, anaphylaxis, angioedema • Hyperglycemia • Injection site erythema Administer only in health care settings where clinicians have access to medications for severe infusion reactions or hypersensitivity reactions. Drug interactions: • Chloroquine/hydroxychloroquine may decrease therapeutic effectiveness of remdesivir – avoid combination
Molnupiravir	• Skin rash • Hypersensitivity reactions, anaphylaxis, angioedema Not recommended for pregnant or pediatric patients. • Molnupiravir may cause harm to the fetus (animal studies: teratogenicity / fetal loss) • Molnupiravir is not recommended in patients <18 years old due to potential for affects on bone and cartilage growth (seen in animal studies)

Pemivibart: IgG1 antibody

PK	IV, t½ range 28-64 days; single dose, may repeat every 3 months
MOA	Targets the spike protein receptor binding domain of SARS-CoV-2 inhibiting attachment to its human ACE2 receptor on host cells
TU	Prophylaxis of COVID-19 for patients ≥12 years old weighing ≥40 kg with moderate to severe immune compromise who cannot be vaccinated or have inadequate response to vaccination
AE	Anaphylaxis, first and second infusions; infusion reactions
	IgG1 crosses the placenta. Data in human pregnancy is minimal. Animal reproduction studies were not conducted.
	Received US Emergency Use Authorization in March 2024

Questions: You may want to pause the video to think about the answers.

- A 10-year-old child is brought to the pediatrician with abrupt onset of fever, nonproductive cough, and complaints of muscle pains 3 hours ago. Rapid antigen test is positive for the circulating virus. The patient is prescribed an antiviral oral suspension that prevents budding of the viral progeny from host cells. The mother is directed to give the child 10 mL of the medicine by mouth every 12 hours for 5 days (10 doses) with food to reduce stomach upset. What is the drug?
- Oseltamivir – metabolized to active oseltamivir carboxylate. Influenza characteristically begins with an abrupt onset of fever, nonproductive cough and myalgia.
- A 68-year-old male with a history of hypertension and diabetes presents to the family practice clinic with complaint scratchy throat, headache, and muscle aches that started yesterday and have progressively worsened. Home COVID testing was positive. After checking for significant drug-drug interactions, the doctor submits an e-prescription for the preferred oral antiviral agent. What is it?
- Nirmatrelvir with ritonavir, the preferred agent.

Questions: Do you want to pause the video?

- A 14-year-old person (pronouns she/her/hers) with a history of severe asthma has a telemedicine visit with her pulmonologist to report that she was exposed to influenza at school yesterday and at her friend's house today. She reports that she is feeling well and has no symptoms. The patient received her flu vaccine 7 days ago. She had an allergic reaction to the oral NAI last year. An e-prescription is submitted for one dose of an oral antiviral agent having a unique mechanism of action and a long half-life. She is advised not take to it at the same time as dairy products or vitamin and mineral supplement. What is the mechanism of action of the antiviral agent?
- Baloxavir marboxil is converted to active baloxavir, a cap-dependent endonuclease inhibitor that blocks influenza A and B proliferation by inhibiting the host cellular pre-mRNAs that serve as primers for viral mRNA. This action prevents initiation of viral mRNA synthesis.
- A 77-year-old male with dyspnea and a positive COVID-19 test is hospitalized for treatment with an intravenous antiviral agent. What is the mechanism of the drug?
- The drug is remdesivir, the only one of the currently available SARS-CoV-2 antiviral agents for the treatment of hospitalized patients not on a mechanical ventilator. It is an adenosine nucleotide analog triphosphorylated and taken up by RdRp. It causes delayed termination of RNA synthesis.

Summary of SARS-CoV-2 Antiviral Agents

Small molecule drugs that inhibit SARS-CoV-2 replication by different mechanisms

Nirmatrelvir-ritonavir preferred agent	Remdesivir	Molnupiravir
3CLPRO main viral protease inhibitor	RdRp inhibitor Delayed chain termination	Lethal mutagenesis (viral error catastrophe) by uptake NHC-TP
Oral, CYP3A4 Ritonavir booster	IV infusion	Oral
Pts 12yrs; 40kg	Pts 28 days old outpatient or hospitalized	Pts 18yrs; 40kg
BID x5d	Daily x3d or 5d	BID x5d
Nausea, diarrhea Dysgeusia Ritonavir-related	Bradycardia Skin reactions	Potential for fetal harm, bone/cartilage growth in peds <18yo (animal studies)

Nirmatrelvir with ritonavir: Mild to moderate illness; outpatients at risk for severe COVID-19 or hospitalization
Remdesivir: Outpatients with mild to moderate illness at high risk for progression to severe COVID-19 (IV daily x3 days) and hospitalized patients not on mechanical ventilation (IV daily x5 days usually)
Molnupiravir: Mild to moderate illness; outpatients at risk for severe COVID-19 or hospitalization
Pemivibart: Single IV dose for p rophylaxis of COVID-19 for patients ≥12 years old weighing ≥40 kg with moderate to severe immune compromise who cannot be vaccinated or have inadequate response to vaccination
Administration of remedesivir and pemivibart only in health care settings where clinicians have access to medications for severe infusion reactions and hypersensitivity reactions.

Questions and Answers

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https://pacificcross.com.vn/upper-respiratory-tract-infections/

Resources

- Access Medicine Brunton LL, Hilal-Dandan R, Knollmann BC. Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e; 2022; Chapter 62 Antiviral Agents (Nonretroviral),
- Access Medicine Jawetz, Melnick, & Adelberg’s Medical Microbiology, 28e, 2019; Orthomyxoviruses (Influenza Viruses),
- Access Medicine Katzung’s Basic & Clinical Pharmacology, 16e, 2024; Chapter 49: Antiviral Agents
- Access Medicine Ryan KJ. Sherris Medical Microbiology, *8e;* 2022; Chapter 9 Respiratory Viruses,
- CDC: Seasonal influenza > Treatment, reviewed Dec 15, 2022 https://www.cdc.gov/flu/treatment/whatyoushould.htm
- CDC COVID-19 Treatment Guidelines, last updated July 12, 2024 https://www.cdc.gov/covid/treatment/index.html
- IDSA 2018 Clinical Practice Guidelines, Seasonal Influenza https://www.idsociety.org/practice-guideline/influenza/
- UpToDate Lexidrug monographs

Lecture and Lab Feedback Form:

https://www.surveymonkey.com/r/NYITCOM_Feedback_Form